Version: 2.2

Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
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Clinical Practice Procedure

Title of Clinical Procedure **Rectus Sheath Catheters for the management of pain following abdominal surgery**		Version No: 2.2
Author: Dr Stephan Clements, Consultant Anaesthetist		**Pages:** **16**
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Authorised by: **Dr Ian Johnson, Acute Pain Lead. Ysbyty Gwynedd**		

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1. Introduction and Purpose of the Written Control Document

The aim of this document is to provide guidance on the safe and effective use of wound catheters for continuous infusion and intermittent boluses of local anaesthesia, with a particular focus on rectus sheath blocks.

Background

After surgery, local anaesthetic can be injected either near nerves, for example rectus sheath block, wound infiltration or into a joint or body cavity. The anaesthetist or surgeon can extend the effect by inserting a fine bore catheter into the appropriate area to infuse local anaesthesia. This can be via an elastomeric pump device, such as the Medfuser or Pajunk Fuserpump.

Rectus sheath catheters are sited to block the anterior divisions of the T6-T11 thoraco-abdominal intercostal nerves. These nerves pass anteriorly between the transversus abdominis and internal oblique muscles, before passing medially to pierce and supply sensation to the rectus and overlying skin. Therefore a catheter placed anterior to the posterior sheath will block these nerves and achieve reduced pain fibre transmission from a midline laparotomy wound.

2. Scope

This document applies to all adults undergoing surgery requiring an abdominal incision at Ysbyty Gwynedd.

This document is intended for all physicians, nurses and pharmacists involved in the perioperative period.

Continuous wound catheter infusions may also be considered for other wounds such as subcostal abdominal wounds or orthopaedic procedures. Sections of this document may be used a reference for such situations.

3. Responsibilities

3.1.1	Anaesthetist
	Overall responsibility for perioperative analgesia
	Prescribing local anaesthetic infusion
	Preparation or ordering of elastomeric pump
	Provide out of hours advice and support
3.1.2	Surgeon
	Effective insertion of wound catheters and initial LA bolus administration
	Ongoing postoperative monitoring and care on surgical ward
3.1.3	Acute Pain Team
	Advice and support New staff elastomeric pump training
3.1.4	Ward Nursing Staff
	Providing adequate post-operative monitoring of patient.
	Safe administration of local anaesthetic bolus injections and removal of catheters.
3.1.5	Pharmacy
	Preparation of elastomeric pump by hospital aseptic department.

4. Clinical Indications

4.1	As an alternative to epidural analgesia
4.2	As part of multimodal approach to analgesia to reduce opioid requirements and side effects

5. Contraindications

5.1	Patient Refusal
5.2	Known local anaesthetic allergy
5.3	Infection at site of insertion

6. Support and Advice

6.1	For advice between the hours 09:00-16:30 normal working week, contact the pain specialist nurses via bleep 115.
6.2	Out of hours, contact on call anaesthetist on bleep 063.

7. Rectus Sheath Catheter Insertion

7.1	**Aim:** Place an infusion catheter on each side at the superior end of the laparotomy wound.
7.2	**Position:** 5 cm of infusion catheter (tip pointing cranially) within the peritoneum-muscle interface.
7.3	**When:** Just **prior** to peritoneal closure.
7.4	**Equipment:** • 1x Rectus Sheath Catheter set. • 2x Smiths-medical 'Lock-it plus' • 2xTegaderms. • 2x 20ml 0.25% (2.5mg/ml) Levobupivicaine • 1x 10ml 0.25% (2.5mg/ml) Levobupivicaine (Max dose: 2mg/kg)
7.5	**How:**
7.51	Place one hand inside the abdomen and with the other hand, insert the Tuohy needle, through the skin and fascia.
7.52	Feel when the needle tip is just superficial to the interface between the peritoneum and muscle layer (*Fig.1*). Palpate and therefore avoid, the inferior epigastric artery.
7.53	Feed catheter through until a 5cm length is in the peritoneum-muscle interface.
7.54	Remove the needle stylet. Hold onto the catheter tip while the needle is removed.
7.55	Secure to skin with the 'Lock-it plus' **prior** to wound closure. (*Fig.2*)
7.56	Attach catheter connectors and inject 20ml 0.25% Levobupivicaine into each catheter separately.
7.57	Flush the Y-connector with 0.25% Levobupivicaine, then attach.
7.58	Close surgical wound.
7.59	Cover 'Lock-its' with Tegaderm dressings.



Figure 1.



Figure 2.

8. Prescription

8.1	**All** prescriptions are the responsibility of the **anaesthetist**. Chose option 2 only when a pump is unavailable.
8.2	**Option 1: Continuous Infusion**
8.21	Prescribe on ePMA (in 'R*egional infusions*' order set) **0.125% Levobupivicaine 8ml/hr via Elastomeric Pump** Comments: Remove catheters on day 3 post op.
8.3	**Option 2: Bolus Administration (when no pump available)**
8.31	Prescribe on ePMA the following: **Levobupivacaine 25mg/10ml solution for injection ampoules** **15ml bolus 'infiltration' 6-8hrly.** Comments: Via Wound Catheter: Remove catheters before 5 days
8.4	Prescribe regular Paracetamol and Opioid PCA for multimodal analgesia.
8.5	Complete an acute pain team referral form

9. Set up of Elastomeric Medfuser 600ml

9.1	This is the responsibility of the anaesthetist. They must familiarise themselves with the product information sheet.
9.2	The anaesthetist primes the pump with **520ml of ~0.125% Levobupivacaine** **12 x 10ml (120 ml) 0.5% Levobupivacaine + 400ml 0.9% saline** Adhere to a **strict** aseptic technique, in theatre: (mask, sterile gloves). Hang 500ml 0.9% saline bag on a drip stand. Place pump, filler needle, 50ml luerlock syringe, 3-way tap giving set & local anaesthetic on a 90cm-by-90cm sterile drape. Pass giving set spike to assistant (*Fig.3*.). **Important:** Clamp pump line before filling. Use the 50ml syringe to fill pump.
9.3	Clearly label the pump bottle with time, date, patient identification, drug and flow rate. Label pump catheter line with bupivacaine drug label(s). Cover the catheter line/pump attachment with a clear Tegaderm.
9.3	Prescribe on the ePMA as above (**8.21 Continuous infusion)**
9.5	Document on anaesthetic chart.



Figure 3.

10. Set up of Elastomeric Pajunk 350ml Fuserpump

10.1	This is the responsibility of the anaesthetist. They must familiarise themselves with the product information sheet.
10.2	The anaesthetist primes the pump with **330ml of ~0.125% Levobupivacaine** **8 x 10ml (80 ml) 0.5% Levobupivacaine + 250ml 0.9% saline** Adhere to a **strict** aseptic technique, in theatre: (mask, sterile gloves). Hang 500ml 0.9% saline bag on a drip stand. Place pump, filler needle, 50ml luerlock syringe, 3-way tap giving set & local anaesthetic on a 90cm-by-90cm sterile drape. Pass giving set spike to assistant Use the 50ml syringe to fill pump (*Fig.4*.)
10.3	Clearly label the Pajunk Fuserpump bottle with time, date, patient identification, drug and flow rate. Label wound catheter line/green rate adjuster box with bupivacaine drug label(s).
10.4	When ready to attach to the wound catheter line, set flow rate at 8ml/hour using the spanner (attached to the back of the flow rate adjuster box) (*Fig*.5). Wrap the attachment with a clear Tegaderm to minimise disconnection
10.5	Document on anaesthetic chart.



Figure 4



Figure 5: To adjust the rate, use the spanner. Located on the reverse of this box

11. Bolus Administration via Rectus Sheath Catheters

11.1	Boluses should be delivered by doctors competent in the use of rectus sheath catheters and in the treatment of local anaesthetic toxicity (Appendix 2). Qualified nurses who have attained competency with the BCUHB IV drug administration and who have had additional training, assessment and have been deemed competent in bolusing these catheters, may also administer a bolus.
11.2	For advice between the hours of 0830 – 1630 contact Pain Specialist Nurse via bleep 115. Out of these hours contact the on-call anaesthetist 063.
11.3	Prepare drug for administration according to prescription, following BCUHB's infection control guidelines.
11.4	Check patient's wrist band against drug chart. Check for allergies.
11.5	Ensure the patient has patent IV access.
11.6	Check the site for signs of migration, leakage and infection. Do not inject if there are concerns re: migration or infection – contact the patient's surgical team, Pain specialist nurse or anaesthetist.
11.7	Non-sterile gloves must be worn and the injection carried out using an aseptic non-touch technique as per infection control guidelines.
11.8	Ensure a filter is in situ.
11.9	Wipe the cap with an alcohol and chlorhexidine swab. Allow this to dry for 30 seconds and remove.
11.10	Attach the syringe and aspirate for blood using a low force.
11.11	If blood is present, **do not** administer the bolus. Inform the surgical team, pain specialist nurse or anaesthetist.

11.12	If no blood is aspirated, inject 5ml of 0.25% levobupivicaine over 2-3 minutes. Wait 5 minutes.
11.13	Ask the patient to inform you of any double vision, tinnitus, numb mouth or metallic taste. Observe for any twitching of the limbs or sudden confusion. If the patient exhibits no side effects, inject the rest of the 0.25% levobupivacaine over 5 minutes. Remove the syringe and replace a sterile cap.
11.15	Monitor the patient for a further 5 minutes for signs of local anaesthetic toxicity.

12. Ward Monitoring and Management

12.1	Close monitoring is essential not only to detect potential complications from the surgery, but also any adverse signs from the local anaesthetic drugs used.
12.2	Observations should be recorded on the patient observation and Early Warning Score tool. After administering a bolus, pulse, blood pressure, respiratory rate, sedation, pain score should be recorded: • Every 15 minutes for 1 hour • Every 30 minutes for 2 hours • Hourly for 4 hours • 4 hourly thereafter or unless the patient's condition indicate otherwise or following bolus administration.
12.3	Intravenous access must be maintained while the wound catheters are being utilised.
12.4	The catheter insertion sites must be inspected everyday for signs of infection or leakage. Report any problems to the acute pain team or on call anaesthetist (out of hours).
12.5	**Removal of catheter**
12.51	The catheters must be removed no later than day 5 or sooner if infection or leakage occurs.
12.52	Any qualified nurse/doctor can remove a local anaesthetic infusion catheter, once a decision has been made to discontinue the infusion.
12.53	Anticoagulation is not a contra-indication to its removal.
12.54	Using an aseptic non-touch technique remove the dressing. Apply gentle traction to the catheter. This should be enough to remove it. If there is any resistance inform the surgical team.
12.55	Ensure the blue tip is intact on the end of each catheter – document in patient's nursing evaluation.
12.56	Cover with a non-occlusive dressing. Remove the dressing after 24 hours.
12.57	If infection is suspected, send the tip for MC&S.

13. Local Anaesthetic Toxicity

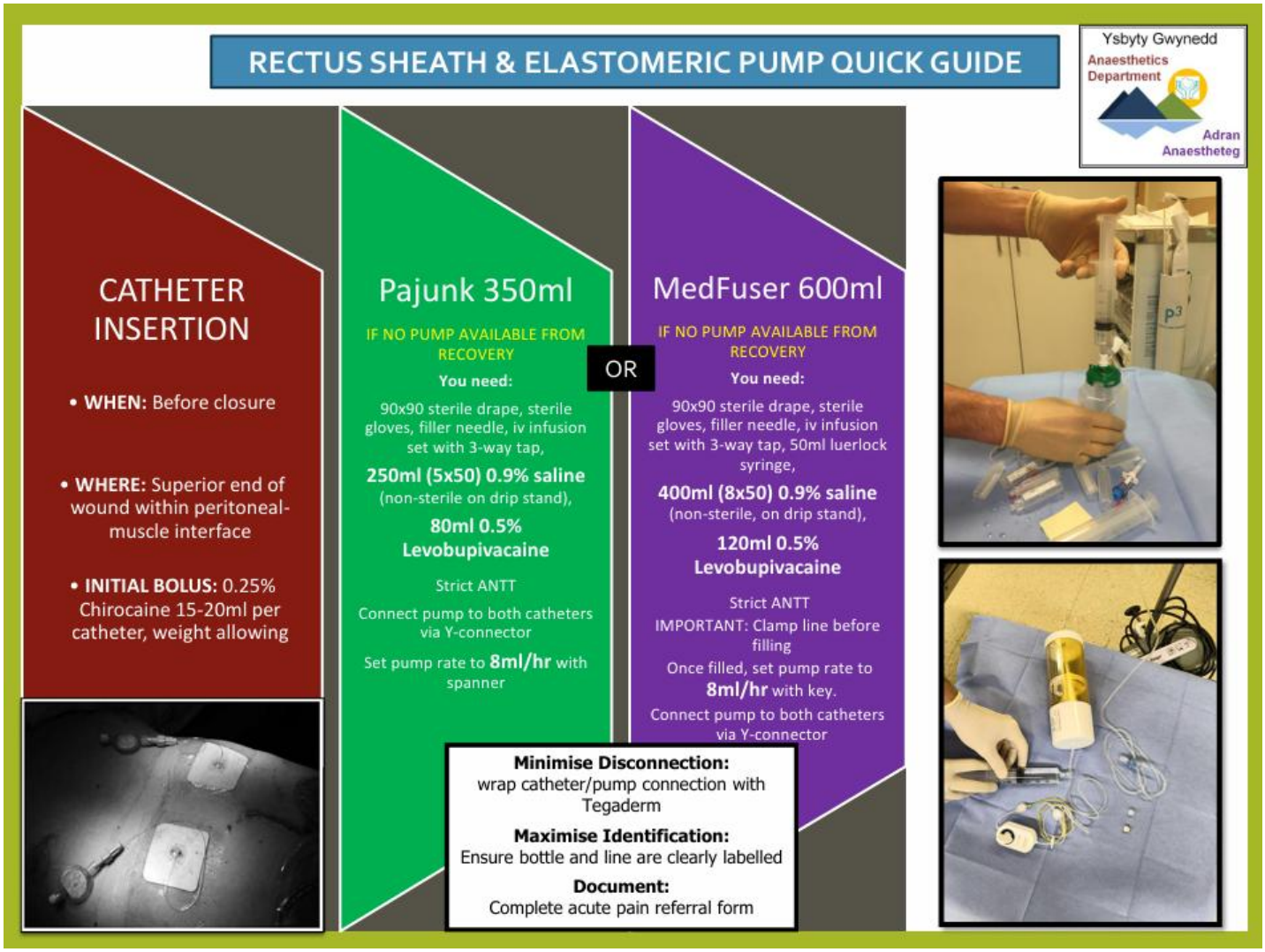
13.1	At low concentrations, local anaesthetic toxicity is rare, but can occur. It is important that the signs are recognised.
13.2	Signs of Toxicity & Treatment
13.3	**Mild:** • circumoral tingling • change in hearing/vision • dizziness, restlessness, confusion
13.4	**Severe:** • sudden alteration in mental status, loss of consciousness, seizures • cardiovascular collapse; bradycardia, hypotension, cardiac arrest
13.5	**Treatment:** • Stop local anaesthetic administration- clamp pump or stop injection. • Call surgical team or 2222 urgently as appropriate. • Continue as per AAGBI safety guideline for management of severe local anaesthetic toxicity (Appendix 2)
13.6	**Your nearest Intralipid bag is Theatre Recovery**

14. References

1.AAGBI (2023). Management of severe local anaesthetic toxicity. {Online} LAST 2023.
2. Crosbie, E. et al. The surgical rectus sheath block for post-operative analgesia:a modern approach to an established technique. European journal of obstetrics and gynaecology and reproductive biology. 2012. 160:196-200.
3. Frank McDermott et al. Surgically placed rectus sheath catheters. Update in Anaesthesia 2010;26:9-11.
4. Gupta, A et al. A meta-analysis of the efficacy of wound catheters for post-operative pain management. Acta Anaesthesiologica Scandinavica. 2011. 55:785-796.

15. Appendix

1.1: Quickguide



1.2: Local Anaesthetic Toxicity

3-10 Local anaesthetic toxicity v.2

Signs of severe toxicity:
- Sudden alteration in mental status, severe agitation or loss of consciousness, with or without tonic-clonic convulsions.
- Cardiovascular collapse: sinus bradycardia, conduction blocks, asystole and ventricular tachyarrhythmias may all occur.
- Local anaesthetic toxicity may occur some time after an initial injection.

START

❶ Stop injecting the local anaesthetic (remember infusion pumps).

❷ Call for help and inform immediate clinical team of problem.

❸ Call for cardiac arrest trolley and lipid rescue pack.

❹ Give 100% oxygen and ensure adequate lung ventilation:

- Maintain the airway and if necessary secure it with a tracheal tube.
- Avoid hypercarbia – consider mild hyperventilation.

❺ Confirm or establish intravenous access.

❻ **If circulatory arrest:**

- Start continuous CPR using standard protocols (→ **2-1**) **but**:
- **Give** intravenous lipid emulsion (Box A).
- **Use smaller adrenaline dose** (≤ **1µg.kg^{-1}** instead of 1 mg)
- Avoid vasopressin.
- Recovery may take >1 hour.
- Consider the use of cardiopulmonary bypass if available.

If no circulatory arrest:

- Conventional therapies to treat hypotension, brady- and tachyarrhythmia.
- **Consider** intravenous lipid emulsion (Box A).

❼ Control seizures:

- Small incremental dose of benzodiazepine is drug of choice.
- Thiopental or propofol can be used, but beware negative inotropic effect.
- Consider neuromuscular blockade if seizures cannot be controlled.

Box A: LIPID EMULSION REGIME

USE 20% Intralipid® (propofol is not a suitable substitute)

Immediately
- Give an initial i.v. bolus of lipid emulsion 1.5 ml.kg^{-1} over 2-3 min (~100 ml for a 70 kg adult)
- Start an i.v. infusion of lipid emulsion at 15 ml.kg^{-1}.h^{-1} (17.5 ml.min^{-1} for a 70 kg adult)

At 5 and 10 minutes:
- Give a repeat bolus (same dose) if:
  - cardiovascular stability has not been restored or
  - an adequate circulation deteriorates

At any time after 5 minutes:
- Double the rate to 30 ml.kg^{-1}.h^{-1} if:
  - cardiovascular stability has not been restored or
  - an adequate circulation deteriorates

Do not exceed maximum cumulative dose 12 ml.kg^{-1} (70 kg: 840 ml)

Box B: CRITICAL CHANGES

Cardiac arrest → Check already done ❶ to ❺, then → ❻

Box C: AFTER THE EVENT

Arrange safe transfer to appropriate clinical area
Exclude pancreatitis: regular clinical review, daily amylase or lipase
Report case on your local critical incident system and to the relevant national system (these vary between each devolved nation and in Ireland)

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